HEXIS FORENSICS

User Manual

Extraction and preservation of digital evidence, step by step

HexisForensics 1.0.0
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Guided by the ISO/IEC 27037 standard

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Note — the program is in Portuguese

Background. The **program** and the **documents it generates** (reports and the expert report) are in **Portuguese**. This English manual explains how to operate it; the names of buttons and screens are quoted in Portuguese, **as they appear in the interface**, so you can find them without hesitation. The module tables in chapters 4, 6 and 8 also show the names and descriptions in Portuguese, because they are read from the program itself.

Notice — personal data processing

Warning. Extracting and examining data from a device is **personal data processing**. Use the tool only with a **legal basis** (a court order, a request from a competent authority, or the documented consent of the data subject) and in accordance with the rules applicable in the country where the examination is performed. Collect only what is necessary and store it securely. The lawfulness of the examination is the responsibility of whoever performs it and whoever authorizes it.

01 Introduction

Hexis Forensics is a free tool for the extraction and preservation of digital evidence from Android phones, iPhones and storage media. It was built for routine work: collect what answers the examination, record how each piece of data was obtained, and deliver documents that stand on their own.

Why this tool exists

Much of the examination work on phones and media is routine and does not need advanced features. Even so, those who do this work usually run into two bad roads.

Paid tools are expensive for routine work

- **Expensive license.** Annual, per-station licenses priced in dollars make small-scale use unfeasible.
- **Idle capacity.** You pay for advanced features that simple, recurring cases never use.
- **Expensive training.** Certified training is costly, when most cases call for solid, elementary training.
- **Moving the device.** The evidence travels to wherever the tool is: more steps in the chain of custody and more risk to volatile data.

Free tools are free, but not simple

- **Hard to operate.** Command line, technical environment and little documentation: not very friendly for the ordinary IT user.
- **No forensic documentation.** In general they do not produce a chain of custody or an expert report: the examiner builds all of that separately.

In practice, the free tool is cheap to download and expensive to learn: the complexity of operating it turns into the complexity and cost of training.

> **Background.** What was missing: a tool that is simple, documented and free of license cost for routine acquisition.

The proposal

Routine logical acquisition, simple to operate, with chain of custody, hash sealing and documents. Each pain has an answer:

Pain	Answer
Expensive license	No license cost.
Idle capacity	Focus on what routine needs: Android, iPhone and media.
Expensive training	Guided operation, screen by screen, within reach of solid, elementary training.
Moving the device	Windows 10 or 11 and a USB cable, wherever the device is.
Complexity of free tools	Graphical interface, no command line.
No forensic documentation	Chain of custody, hash sealing and expert report already included.

What it does, and how far it goes

The reach of each collection depends on the platform and the type of extraction. The table summarizes what version 1.0 does:

Platform	Logical	File system	Physical (bit by bit)
Android	Yes. On a common device, with no elevated privilege.	Partial. Without root, only the shared storage: the apps' databases do not come.	Only with root. Data partition, DES profile. Without root, it fails explicitly.
iPhone	Yes. Common and encrypted backup.	No. It would require jailbreak.	No. It would require exploiting the device's boot.
USB drives and disks	Yes. Copy of the user's folders and an index with hash.	Partial. Index of the whole drive (path, dates and SHA-256), without copying the content; the copy reaches only the user's folders.	Yes. .img image of the volume; requires Administrator and the DES profile.

Focus: routine logical acquisition. Full file system and physical image only with elevated privilege, never on the iPhone, and declared as such.

How it works, in five steps

1. **Connect.** Android over USB debugging, iPhone paired, media with a write blocker.
2. **Declare the method.** The Method and limitations screen states what the collection does not reach.
3. **Identify the item.** Case, authorization, make and model, seal and two photographs.
4. **Extract with hash.** A manifest with the SHA-256 of each file: the seal of the acquisition.
5. **Generate the documents.** Reports, selected conversations and the expert report.

The documents it generates

Each acquisition generates documents that stand on their own:

- **Extraction Report.** Timeline of the procedure, environment, modules, outcome and the hash manifest.
- **Content Report.** Messages, calls and contacts interpreted, with the origin of each line.
- **Selected conversations.** In PDF, only the relevant conversations, complete.
- **Expert Report.** The final document, signed by the examiner of the session.

All of them are written **outside** the sealed acquisition, which is never altered (chapter 9).

The differentiator: not collected is not does-not-exist

Each module ends in one of three states:

State	What it means
Completed	The module obtained what it set out to.
Completed with a caveat	It obtained part, and declared what it did not reach.
Obtained no data	None of that material was acquired, which does not mean the data does not exist on the device.

In practice:

- **Query refused by the system:** the refusal is preserved and declared; an empty response is recorded as such.
- **Encrypted backup not decrypted:** "nothing is asserted" about the content.
- **Locked device:** the report says why it did not extract.

Why it matters: the absence of a piece of data in the material is never presented as the absence of that data on the device.

The tool's principles

Three ideas guide every screen and every document. Knowing them helps you read what the tool says — and what it, on purpose, does not say.

- **What is necessary, well recorded.** The tool does not measure quality by volume. It collects what matters to the examination and documents the provenance of each piece of data: which file it came from, by which route, with which hash.
- **Never assert what was not verified.** When a piece of data is not obtained, the document says WHY — the system refused, the module was not run, the backup did not contain it. The absence of data in the material is never presented as the absence of data on the device.
- **Every change to the device is recorded and justified.** The modules that touch the device are flagged, start unchecked and leave in the procedure record what was done and why.

What the tool does not do

Being honest about its reach is part of the method. The tool does **not** unlock a device, does not bypass a passcode, does not recover deleted data, does not extract from the cloud, does not perform physical extraction of an iPhone, and, on Android, only performs physical extraction with root. It also does not decode every existing app. For those cases there are specialized tools. Hexis Forensics covers routine logical acquisition — and what surrounds the extraction: chain of custody, hash sealing, expert report, examination without extraction and documented delivery.

The DEFR and DES roles

The ISO/IEC 27037 standard distinguishes the **DEFR** (first responder: identifies, collects and preserves following a procedure) from the **DES** (specialist: decides when something falls outside the procedure). Each module declares the minimum profile it requires. The tool stops no one — it informs, and the report records it.

Compliance and transparency

Compliance

- **ISO/IEC 27037.** Identification, collection, acquisition and preservation.
- **Chain of custody.** Seal, photographs and a record of each step, in accordance with the law applicable in each country.

Transparency

- **Installer with a published hash.** The SHA-256 lets you verify the file before installing (chapter 2).
- **Tested on real devices.** Motorola, Samsung, iPhone 11 and a USB drive, with what was proven on each one (chapter 11).
- **Declared limitations.** For example: the agent not verified on Xiaomi; the IMEI not delivered without root from Android 10 on.

Going further: the Official Training

This manual teaches you to operate the tool. The **Official Hexis Forensics Training** goes further: it teaches the digital forensics that underpins each screen — chain of custody, integrity by hash, the limits of each type of extraction and the writing of the expert report. There is also an offering in partnership with higher-education institutions, and deployment with team training for police and forensic agencies.

02 Installation and first access

From the installer to the first session: requirements, creating your examiner profile and the recovery code, which is the only way back if you forget the password.

Requirements

- A computer with **Windows 10 or 11**.
- A **USB data cable** (some cables only charge and will not work).
- For **iPhone**: the **Apple Devices** app (Microsoft Store) or iTunes, which install the service that communicates with the device.
- For **computer disks**: a USB dock or adapter and a hardware **write blocker** (see chapter 8).

Verify the installer before installing

Along with the installer, its **SHA-256 hash** is published — the fingerprint of the file. Verifying it guarantees that the downloaded file is exactly the one published, without corruption or tampering. It is the same logic as the seal of the acquisitions.

1. In the folder where the installer is, right-click an empty space and choose **Open in Terminal**.
2. Type **Get-FileHash .\HexisForensics-Setup-1.0.0.exe** and press Enter.
3. Compare the value in the **Hash** column with the published one. A single different character means a different file: do not install.

Installation

Run the installer and follow the screens. Check the option **Criar atalho na Área de Trabalho** (create a Desktop shortcut). Cases are kept in the **HexisForensics-Casos** folder, inside your Windows user folder — outside the program folder, so that an update never reaches the evidence. To update, just install the new version over the old one: cases and examiners are preserved.

The blue Windows warning

The installer is free and has no paid digital signature. For that reason, on the first run, Windows shows the **"Windows protected your PC"** screen, with an unknown publisher. It is a warning, not a virus diagnosis:

1. Click **More info**.
2. Check the file name and click **Run anyway**.

Background. If **Smart App Control** on Windows 11 is on, it blocks the installer without offering the button: it requires a digital signature, which this version does not have. There is no portable version. The ways out are to install on another computer, ask your IT department to allow the program, or **run from source** (next section).

Alternative: run from source

When the installer is blocked and there is no way to allow it, the tool runs directly from source, like any open-source project — there is no installation, the same suite is run by Python. It takes a few more steps, but it does not depend on a digital signature.

1. Install **Python 3.13** from **python.org** (under Downloads, choose the Windows installer for version **3.13**). **Do not use 3.14**: one of the iPhone libraries does not yet have a ready version for it. On the first screen of the Python installer, check **Add python.exe to PATH**.
2. Download the code from the project's **Releases page**: **github.com/HexisForensics/HexisForensics/releases**. On the most recent version, download **Fonte (ZIP)** (the source ZIP). Do **not** use **Code** → **Download ZIP** from the main page — that brings the development version, with no tagged release.
3. **Verify the ZIP's hash**, as with the installer: in the file's folder, **Open in Terminal** and **Get-FileHash .\HexisForensics-1.0.0-fonte.zip**. The value must be identical to the one published in the Release.
4. Extract the folder. Open it, right-click an empty space and choose **Open in Terminal**.
5. Type **py -m pip install -r requirements.txt** and press Enter. Wait for the libraries to install (it takes a while the first time).
6. Type **py main.py** and press Enter. The tool opens on the entry screen, just like the installed version.

Background. Running from source, cases are kept in **HexisForensics-Casos**, in your Windows user folder — the same as the installed version, and **outside** the program folder, so that updating (downloading the new ZIP) never reaches the acquisitions. The **adb.exe** already comes in the ZIP, so Android extraction works with no extra step; for the iPhone, the Apple Devices app still applies (Requirements).

First access: your examiner profile

The tool opens on the **entry screen**. Each examiner has their own profile and password: that is what identifies who performed each extraction and who signs each expert report.

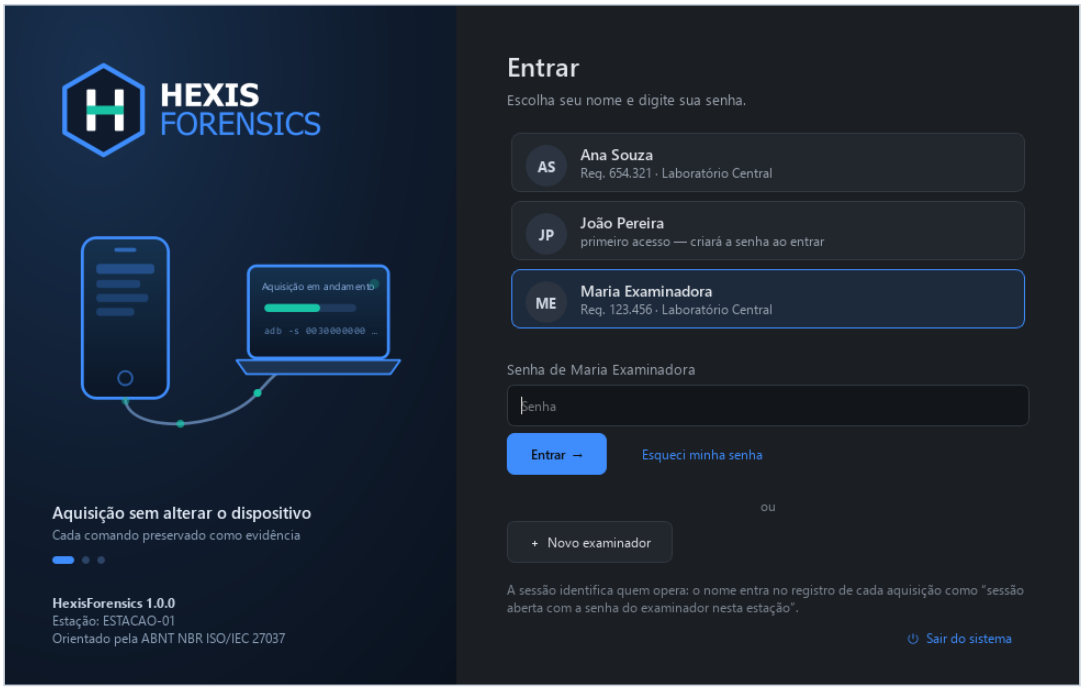


Figure 1 — Entry screen: choose your name and type your password.

1. The first time, click **Novo examinador** (New examiner).
2. Fill in the **name** as it should appear in the documents, the **professional registration** and the **institution or laboratory**.
3. Create a **password** of at least 6 characters and repeat it.
4. Click **Cadastrar e entrar** (Register and enter).
5. Write down or print the **recovery code** (next section), check the confirmation and click **Continuar** (Continue).



Figure 2 — Registration of a new examiner.

The recovery code

When you create the password, the tool shows a 16-character code, in four groups. It appears **only once**. It is the only way to get in if you forget the password: no one — not even the maker — can reset it. Once used, the code is replaced by another, which also appears once.

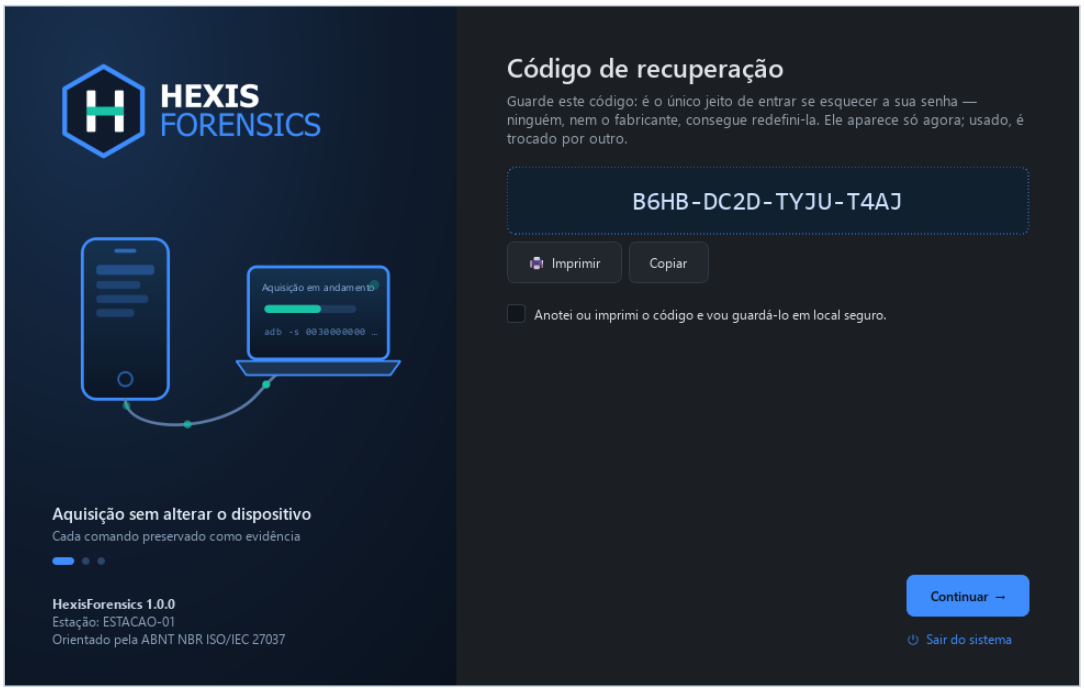


Figure 3 — The recovery code, with the print and copy options.

Warning. Keep the code away from the computer: printed in a folder, in a password vault, in a sealed envelope. Whoever loses the password **and** the code must create a new profile. The old acquisitions stay under their name, because the name is recorded inside the seal of each one.

I forgot my password

1. On the entry screen, choose your name and click **Esqueci minha senha** (I forgot my password).
2. Type the recovery code (with or without dashes, upper or lower case).
3. Choose the new password and click **Redefinir e entrar** (Reset and enter).
4. Write down the **new** code that appears: the old one is no longer valid.

During the session

- Your name appears in the **footer of the side bar**, with the time you entered.
- **Trocar examinador** (Switch examiner) returns to the entry screen, for someone else to log in. You cannot switch during an extraction.
- After **15 minutes of inactivity**, the screen locks and asks for the password again. An extraction in progress **keeps running** in the background.
- Five wrong passwords in a row lock that examiner for 10 minutes.
- In **Examinadores** (Examiners, in the side bar) you correct your data, change the password, generate a new recovery code or delete your profile. The **Novo examinador** button opens the registration of another person: they create their own password and receive their own code, and become the session's examiner. No one sees or alters another examiner's data.
- Logins, lockouts, recoveries and deletions are kept in the station's **access log**.

Background. The record of each acquisition says: “session opened with the examiner's password on this station”. The password guarantees **authorship** — no one operates in someone else's name by carelessness. It does not prove identity (a shared password gets in too) and does not protect the case files, which sit on the disk.

03 Preparing the Android phone

Without elevated privilege, Android extraction is done through ADB, Android's own debugging tool. For that, the device has to authorize the computer.

Step 1 — Enable the Developer options

1. Open **Settings** → **About phone** (on Motorola, **Settings** → **System** → **About device**; on others: "Software information").
2. Scroll to the bottom and tap **7 times** on **Build number** (on some devices: "MIUI version", "OS version" or "Version number"). The device may ask for the unlock code.
3. The message **"You are now a developer"** appears — or, if it was already on, "You are already a developer".



Figure 4 — Build number: tap 7 times. (IMEI and addresses covered in this manual.)



Figure 5 — The developer-mode confirmation.

Step 2 — Turn on USB debugging

1. Open **Settings** → **System** → **Advanced** → **Developer options** (on some devices, under "Additional settings").
2. Turn on **USB debugging**.
3. Also turn on **Stay awake**: the screen does not go off while the device is on the cable, and the connection does not drop in the middle of the extraction.

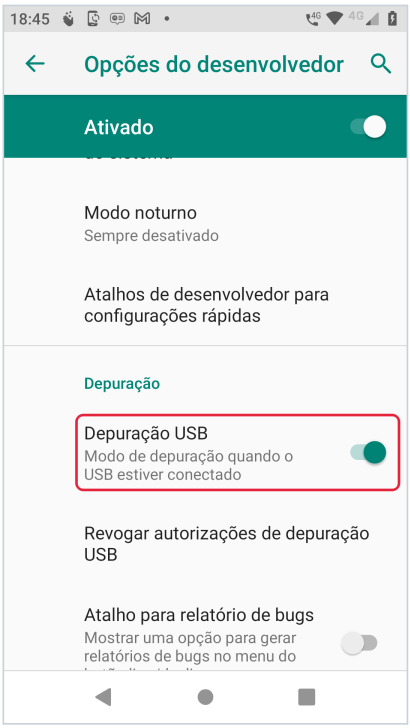


Figure 6 — USB debugging on.

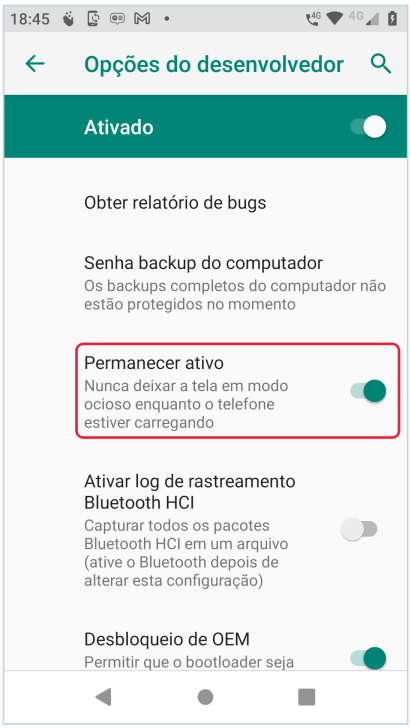


Figure 7 — Stay awake on.

Step 3 — Authorize the computer

1. Connect the data cable. On the device's screen, **Allow USB debugging?** appears.
2. Check **Always allow from this computer**.
3. Tap **Allow**.
4. In the tool, on the **Android** page, click **Verificar Conexão** (Check connection). The status should say the device is connected and authorized.



Figure 8 — The authorization prompt (reproduction: Android does not allow capturing this screen).

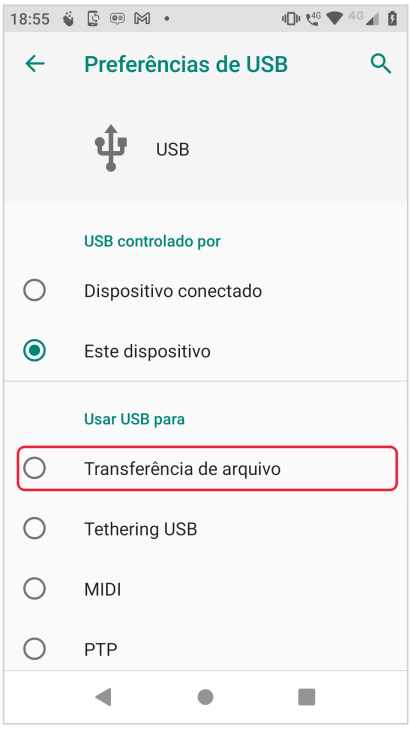


Figure 9 — USB preferences, if the connection drops.

Tip. If the prompt does not appear, or the device shows as **not authorized**: unlock the screen, tap **Revoke USB debugging authorizations** (in Developer options) and reconnect the cable. If the connection drops during the extraction, open **Settings** → **Connected devices** → **USB** and choose **File transfer**.

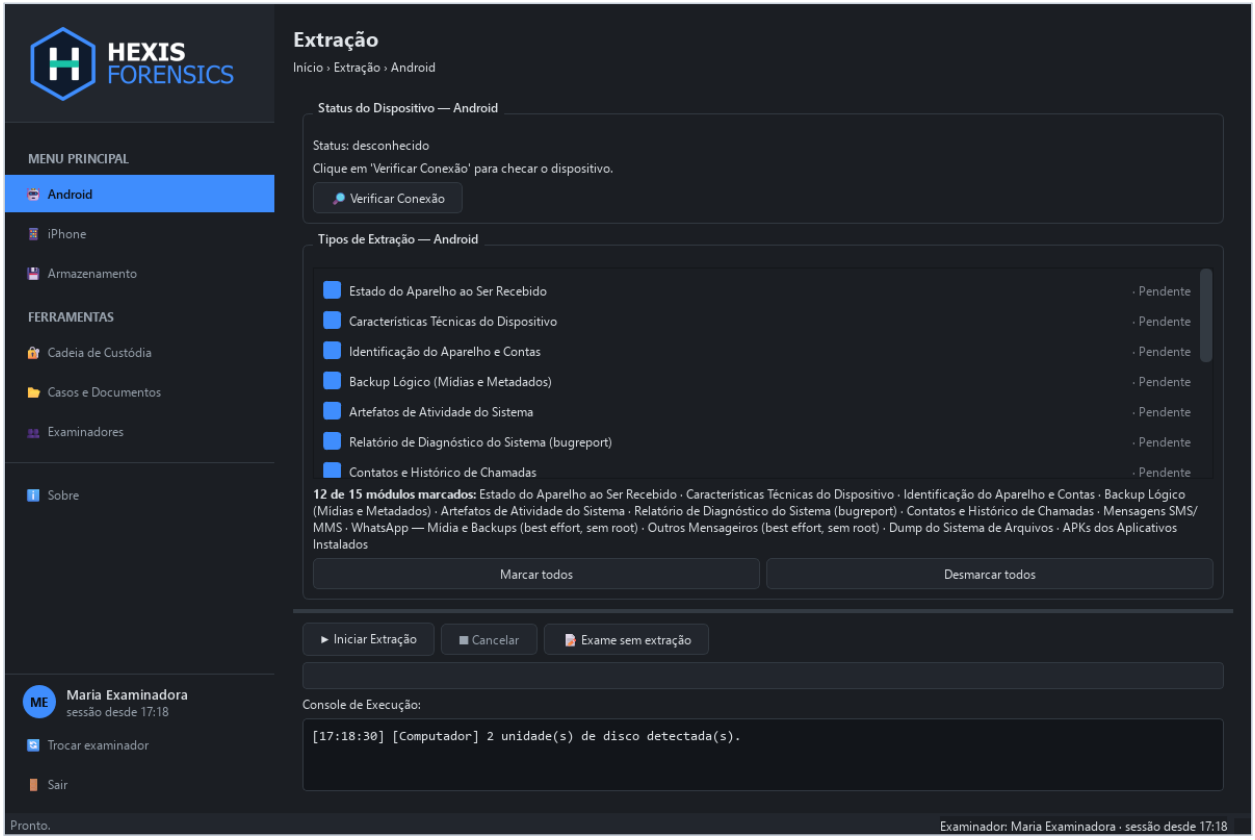


Figure 10 — Android page: device status, extraction modules and console.

Manufacturer specifics

Xiaomi (MIUI / HyperOS)

Besides USB debugging, turn on **Install via USB** and **USB debugging (security settings)**. Both require a **Xiaomi account** signed in and, sometimes, an inserted SIM. Without them, the device refuses the agent app. *In this version, the agent has not yet been verified on a Xiaomi device: if the install is refused, the tool records the refusal and concludes nothing about the content.*

Samsung

The **Developer options** appear at the end of the Settings list, right below **About phone**, after the 7 taps on **Software information** → **Build number**.

Verified on a **Galaxy A22 with Android 13 (One UI 5.1)**, factory settings and with no elevated privilege: the agent app was installed, collected and removed **with no on-screen confirmation** — no “Install app?”, no Play Protect, no permission request. Samsung also delivers the contact list to the common module, without the agent.

On other models or versions, **Play Protect** may ask whether the app is safe. If the question appears, confirm the install on the device's screen.

Warning. Turning on USB debugging and authorizing the computer are changes to the device's configuration. The tool records them in the procedure as a requirement of the method: without them there is no logical acquisition on a device with no elevated privilege.

04 Android extraction, step by step

From clicking Iniciar Extração to the sealing of the acquisition. Each step exists to record something the expert report will need.

Choosing the modules

The list of modules appears on the Android page. The common modules come already checked. Those that alter the device in a relevant way are **express selection**: they start unchecked and the **Marcar todos** (Check all) button does not reach them — they must be chosen one by one, on purpose.

The sequence of screens

1. Click **Iniciar Extração** (Start Extraction). The tool asks for confirmation of the session's examiner ("This extraction will be recorded as performed by…").
2. **Method and limitations**: what the chosen collection will not reach and what may be altered on the device. This is the moment to decide — afterward it becomes a record. Read it and click **Ciente — Prosseguir** (Understood — Continue).
3. **Case**: create a new case or choose an existing one. A case gathers several devices; each device becomes an acquisition within it.
4. **Authorization**: attach the document that authorizes access to the data. Continuing without attaching is allowed, but it is recorded that there was no document in sight.
5. **Device and chain of custody**: make and model, color, condition, **seal number** (required) and at least **two photographs** of the item (front and back).
6. The extraction begins. Follow it through the console and the status column of each module.

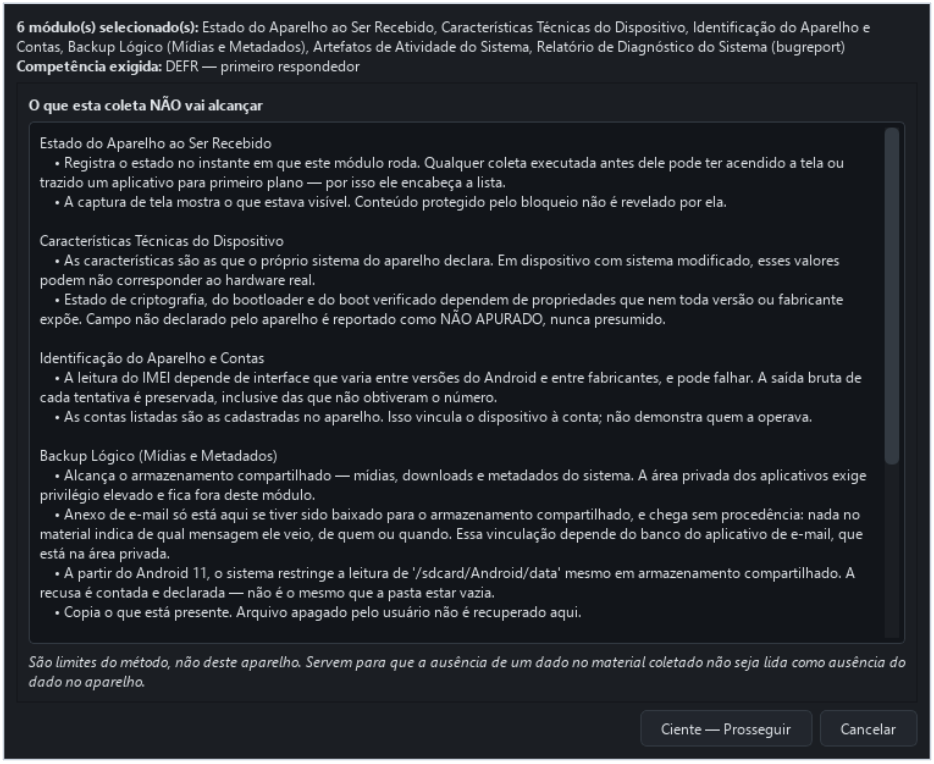


Figure 11 — Method and limitations: what the collection does not reach, before starting.

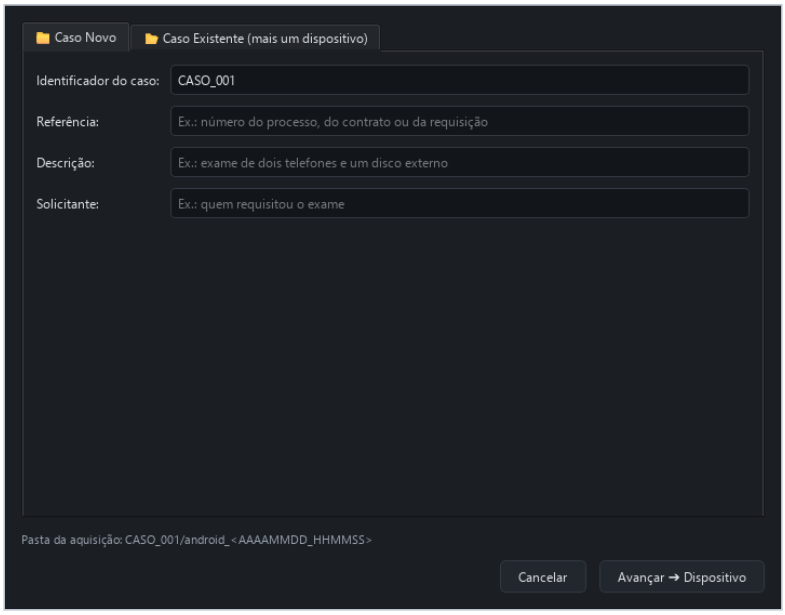


Figure 12 — New or existing case.



Figure 13 — Item, seal and photographs.

How to read the result

Each module ends in one of three states, and they are not interchangeable:

State	Means
Completed	The module obtained what it set out to.
Completed with a caveat	It obtained part, and declared what it did not reach (for example, a query refused by the system).
Obtained no data	None of that material was acquired. This does **not** mean the data does not exist on the device.

At the end, the tool generates the **hash manifest** (the seal of the acquisition): the SHA-256 of each acquired file. It also checks whether the device connected at the end is the same as at the start.

> **Warning.** Do not close the tool during an extraction: the acquisition would be left without a seal. If you need to stop, use **Cancelar** (Cancel) and wait for the extraction-finished message.

The available modules

The table below is generated from the tool's own code; the module names and descriptions appear in Portuguese, as in the program.

Module	What it does	Alters the device?	Requires
Estado do Aparelho ao Ser Recebido	Registra o aparelho como foi recebido: captura de tela, estado de energia e de bloqueio, nível de bateria, aplicativo em primeiro plano e perfis de usuário. Execute PRIMEIRO: qualquer outra coleta pode acender a tela ou trazer um aplicativo para frente, e o estado registrado depois disso já não é o estado em que o dispositivo chegou.	no	profile DEFR
Características Técnicas do Dispositivo	Lê as propriedades do sistema (getprop) e gera um relatório com fabricante, modelo, série, versão do sistema, patch de segurança, estado da criptografia, do bootloader e do boot verificado, além de kernel, armazenamento e bateria. A saída bruta dos comandos é preservada ao lado do relatório.	no	profile DEFR
Identificação do Aparelho e Contas	Coleta o IMEI (por múltiplas vias, com validação do dígito verificador) e as contas cadastradas no aparelho — Google, WhatsApp, Telegram e demais —, além de dados de telefonia e SIM. É o que vincula o dispositivo físico a uma pessoa. LIMITAÇÃO: em Android 10+ a leitura do IMEI pelo shell pode ser recusada pelo sistema; a tentativa e sua recusa ficam registradas como evidência.	no	profile DEFR
Backup Lógico (Mídias e Metadados)	Extrai as pastas DCIM, Download, Pictures, Movies, Music e a área de dados de aplicativos em armazenamento compartilhado, via 'adb pull', além de metadados do sistema (getprop) e da lista de pacotes instalados.	no	profile DEFR
Artefatos de Atividade do Sistema	Coleta a linha do tempo de uso de aplicativos (usagestats), redes Wi-Fi conhecidas, notificações, histórico de bateria, processos, contas configuradas e permissões concedidas por pacote. Responde quando o aparelho foi usado e por qual aplicativo. ATENÇÃO: são dados voláteis — execute este módulo no início do exame, pois reiniciar o aparelho descarta parte do conteúdo.	no	profile DEFR
Relatório de Diagnóstico do Sistema (bugreport)	Gera o bugreport do Android: histórico de energia com marcações de tela ligada, estatísticas de uso por aplicativo, eventos do sistema, conectividade, pacotes com data de instalação e logcat retido. É o artefato mais denso disponível sem root, e o que melhor sustenta uma linha do tempo de uso do aparelho. ATENÇÃO: leva de um a vários minutos e pode ocupar dezenas de MB.	no	profile DEFR

Module	What it does	Alters the device?	Requires
Contatos e Histórico de Chamadas	Consulta os Content Providers nativos do Android (content://call_log/calls e content://com.android.contacts/data/phones). ATENÇÃO: o shell ADB pode não ter permissão READ_CALL_LOG a menos que o dispositivo tenha root; nesse caso o provider retorna 'Permission Denial' (registrado nos arquivos gerados) e o módulo tenta 'adb backup' do app discador como alternativa best-effort em aparelhos com Android abaixo da versão 12.	yes — Quando a leitura direta é negada e a via de backup é tentada, o aparelho exibe pedido de confirmação que precisa ser aceito na tela.	profile DEFR
Mensagens SMS/MMS	Consulta os Content Providers nativos de SMS (content://sms, inbox, sent, draft) e de MMS (content://mms: cabeçalhos, partes, endereços e anexos). ATENÇÃO: o shell ADB pode não ter permissão READ_SMS a menos que o dispositivo tenha root; nesse caso o provider retorna 'Permission Denial' (registrado nos arquivos gerados) e o módulo tenta 'adb backup' como alternativa best-effort em aparelhos com Android abaixo da versão 12.	yes — Quando a leitura direta é negada e a via de backup é tentada, o aparelho exibe pedido de confirmação que precisa ser aceito na tela.	profile DEFR
Agenda, Chamadas e SMS por aplicativo agente (qualquer versão do Android)	Instala no aparelho o aplicativo agente da suíte, que lê agenda, chamadas, SMS e MMS com permissão própria, recolhe o resultado e o desinstala. É a via para Android 12 em diante, onde o backup do sistema não existe. ALTERA O DISPOSITIVO (instala e remove um aplicativo).	yes — Um aplicativo (o agente de coleta da suíte) é instalado no aparelho, recebe as permissões de leitura de contatos, chamadas e SMS, é executado uma vez e é desinstalado ao final.	express selection (starts unchecked); profile DEFR
WhatsApp — Mídia e Backups (best effort, sem root)	Extrai, sem necessidade de root, as pastas de mídia e os backups locais criptografados (.crypt12/.crypt14/.crypt15) do WhatsApp e WhatsApp Business acessíveis no armazenamento externo. LIMITAÇÃO: o banco msgstore.db ativo e a chave de criptografia ficam em /data/data/com.whatsapp e exigem root para acesso.	no	profile DEFR
Outros Mensageiros (best effort, sem root)	Extrai mídia e arquivos em armazenamento externo de Telegram, Signal, Instagram, Messenger, Facebook, Snapchat e Threema, para os que estiverem instalados. LIMITAÇÃO: o banco de mensagens de cada aplicativo fica em área privada (/data/data) e exige root — a ausência de mensagens neste módulo NÃO significa ausência de mensagens no aparelho.	no	profile DEFR
WhatsApp — Mensagens por substituição temporária do aplicativo (sem root)	Obtém o banco de mensagens em claro do WhatsApp em aparelho sem root: substitui temporariamente o aplicativo por uma versão antiga que aceita 'adb backup', tira o backup e reinstala o original. ALTERA O DISPOSITIVO. Exige o APK antigo em 'binario/whatsapp_legado.apk' e Android anterior ao 12.	yes — O aplicativo WhatsApp é desinstalado com preservação dos dados, substituído por uma versão antiga e, ao final, reinstalado na versão original.	express selection (starts unchecked); profile DES

Module	What it does	Alters the device?	Requires
Dump do Sistema de Arquivos	Copia recursivamente /data/data (bancos SQLite dos apps) e /sdcard. Arquivos protegidos serão bloqueados caso o ADB opere em modo comum (User Shell).	no	profile DEFR
APKs dos Aplicativos Instalados	Copia o arquivo .apk de cada aplicativo instalado pelo usuário e calcula seu SHA-256, permitindo identificar aplicativo adulterado ou malicioso e submetê-lo a análise posterior. Aplicativos de sistema não são copiados. ATENÇÃO: pode levar vários minutos e ocupar espaço considerável, conforme a quantidade de aplicativos instalados.	no	profile DEFR
Extração Física (Bit-a-Bit)	Clonagem bit-a-bit da partição 'userdata' via 'dd', transmitida por canal binário e com SHA-256 calculado durante a aquisição. Requer privilégios de root ou hardware legado vulnerável.	no	requer root; profile DES

05 WhatsApp, step by step

Without elevated privilege, the WhatsApp message database sits in a private area of the app, out of reach. The route is WhatsApp's own local backup, protected with a 64-digit key that the examiner knows. The screens in this chapter are from a real device; the key appears covered.

Why this is the route

WhatsApp writes encrypted local backups (**msgstore.db.crypt15**) in a folder the tool can copy. When the backup is protected by a **64-digit key** — and not by a password or a passkey —, whoever has the key can decrypt it. The tool decrypts it, checks the authenticity of the content and reads the conversations.

Warning. Touching the backup's protection and telling WhatsApp to make the backup are **changes to the examined device**, made by the examiner. When it decrypts, the tool records them in the case addendum, together with the key.

Step 1 — See how the backup is protected

In WhatsApp, open **Settings** → **Chats** → **Chat backup** and scroll to **End-to-end encrypted backup**. What is written beneath it decides the route:

What appears	What to do
Off	Go to Step 2.
On · Password or **On · Passkey**	No good: the tool decrypts only with the 64-digit key. Tap the option, tap **Turn off**, enter the backup password and confirm. Then, Step 2.
On · 64-digit key	Good, if someone has the key written down. WhatsApp only shows the key when it is created: without it written down, turn off and create a new one (Step 2).

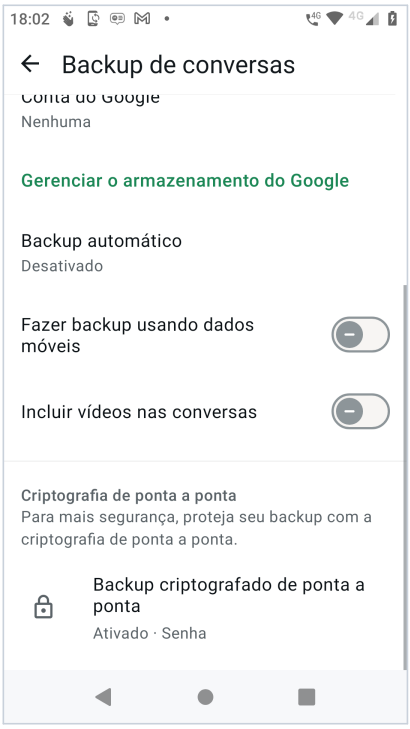


Figure 14 — On · Password: no good.



Figure 15 — Backup protected with a password: tap Turn off.

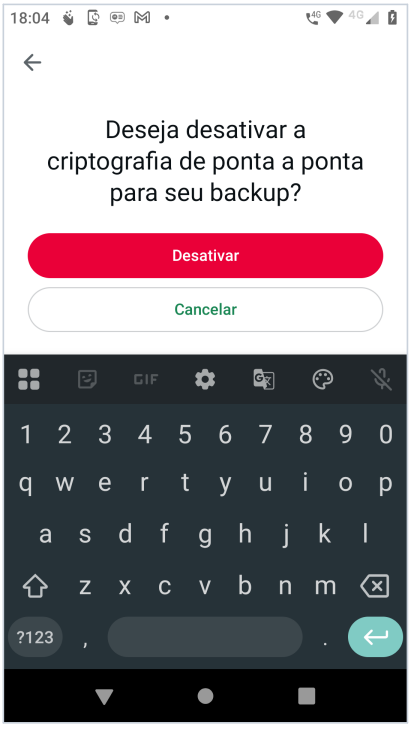


Figure 16 — Confirm turning it off.

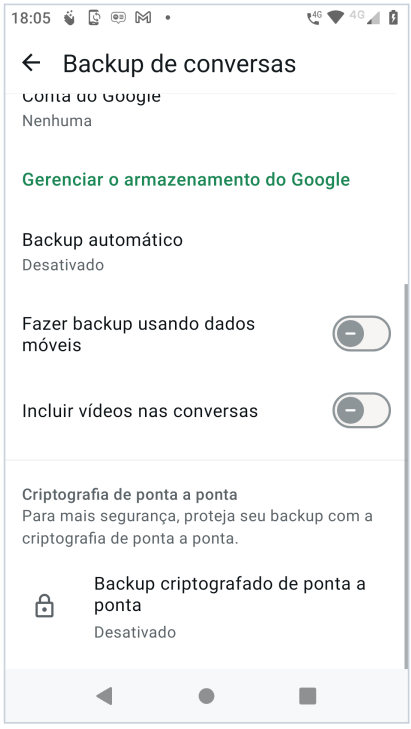


Figure 17 — Now: Off.

Step 2 — Turn it on with the 64-digit key

1. Tap **End-to-end encrypted backup**.
2. **DO NOT tap the green "Use passkey" button.** Tap **More options**.
3. Choose **64-digit encryption key** (not "Password").
4. Tap **Generate your 64-digit key**.
5. The key appears: 16 groups of 4 characters. **Write it down** — or read it over the cable (see tip). Tap **Continue**.
6. WhatsApp shows the key again for checking. Tap **I saved my 64-digit key**.
7. Tap **Create**. The process finishes in the background.



Figure 18 — The trap: the green button is no good. Tap More options.

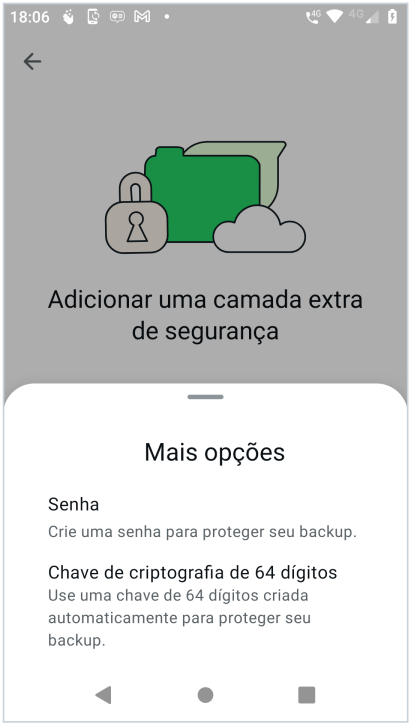


Figure 19 — Choose the 64-digit key.



Figure 20 — Generate the key.



Figure 21 — The key (covered in this manual).



Figure 22 — Check and confirm you saved it.



Figure 23 — Create the protected backup.

Tip. With the cable connected, the key can be read from the screen by ADB, with no typing error: the screen text is obtained with **uiautomator dump**. When in doubt, photograph the key screen and add the photo to the case.

Step 3 — Make the backup ON THE DEVICE

1. In **Chat backup**, check that **Google Account** is **None** and **Back up automatically** is **Off** (in older versions: **Back up to Google Drive** set to **Never**).
2. Tap **Back up** and wait for it to finish.
3. Check the **Local:** line — it should show the current time.



Figure 24 — Protected with the 64-digit key.

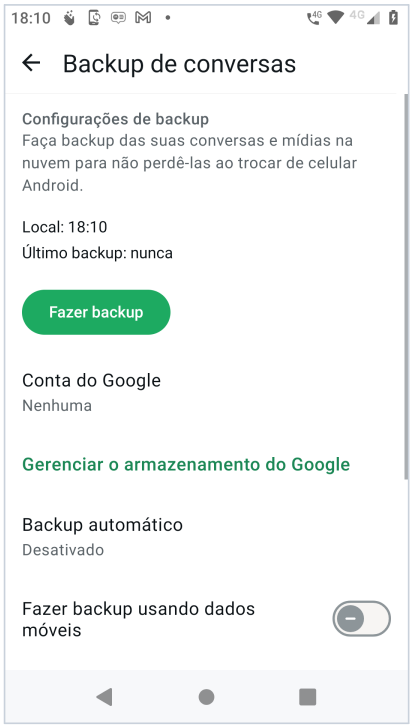


Figure 25 — Local backup done: check the time under "Local".

Warning. Make the backup **on the day of the examination**, after turning on the key. An old backup — or one made with a password — does not open with the new key and does not bring the recent messages.

Step 4 — In the tool: extract

Run the extraction normally (chapter 4), with the module **WhatsApp — Mídia e Backups** checked. It copies the media and the encrypted backups and records, by asking the device itself, **when each backup was written**.

Step 5 — In the tool: decrypt

1. Open **Casos e Documentos** (Cases and Documents), choose the case and the acquisition.
2. Click **Decifrar backup** (Decrypt backup).
3. Type the key into the 16 fields (or paste the whole key into any field).
4. Click **Decifrar** (Decrypt). The decrypted backup goes to the case's **decifrados** folder, **outside** the sealed acquisition, with its provenance alongside. The key is recorded in the case addendum, so that another examiner can repeat the decryption.
5. Next, the **Content Report** is generated, with the conversations.

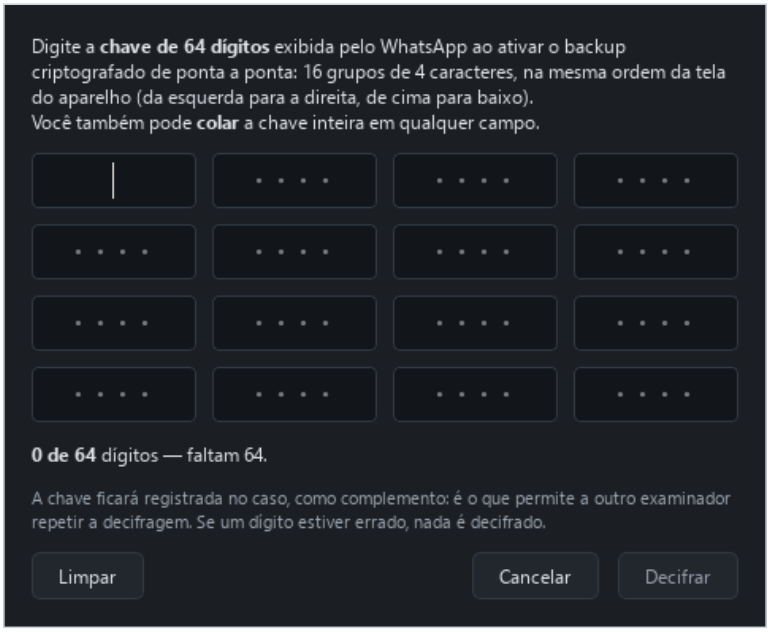


Figure 26 — The 64-digit key, in 16 groups, in the tool.

Step 6 — Read and select the conversations

The Content Report brings the spreadsheet (CSV) and the HTML notebook with the full text. Very long conversations appear in the PDF as an index, so the document does not run to thousands of pages. For the expert report, use **PDF de conversas selecionadas** (PDF of selected conversations): check the relevant conversations and the tool generates the PDF of just those, complete, with the origin of each message.

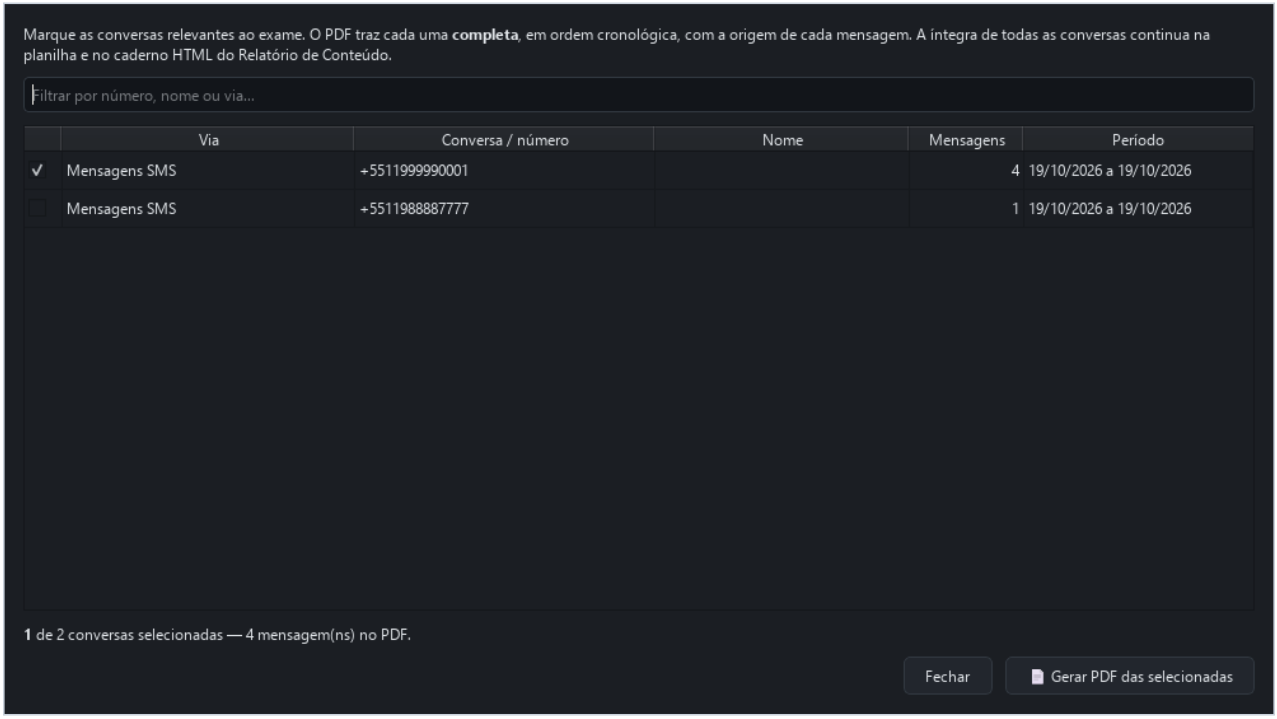


Figure 27 — Choosing the conversations that go into the PDF.

The backup date in the documents

The decrypted content portrays the device **on the date the backup was written**, not on the day of the examination. For that reason the report and the expert report carry the **Source note**: the backup date (reported by the device itself and recorded in the seal), the most recent message it contains and the warning that later messages are not in the material.

When the backup does not open

- **Key typed wrong**: nothing is decrypted. Check it group by group.
- **Backup protected by a password or passkey**: it does not open with any key. Redo steps 1 to 3 and extract again.
- **Backup made before turning on the key**: it does not open with it. Make a new backup (step 3) and extract again.
- **.crypt14 or earlier backup**: it does not open with the 64-digit key.
- **No backup on the device**: the module says it only found media. This does not mean an absence of messages.

06 iPhone

Without jailbreak, the route is the iPhone's local backup — the same one iTunes would make —, obtained after the device trusts the computer.

Before you start: the computer

1. Install the **Apple Devices** app (Microsoft Store) or iTunes. It installs the Windows service that talks to the iPhone. You do not need to open it or sync anything — and **do not** sync: syncing writes to the device.
2. If the iPhone was already on the cable during the install, **disconnect and connect it again**: the service only sees the device connected after it has started running.

Before you start: the iPhone

1. The iPhone must have **finished the initial setup**. On a restored device, iOS refuses the backup until then (the tool shows “initial setup not completed — code 212”). The **Apple account** screen can be skipped: tap **Forgot password or don't have an Apple Account?** and choose to set it up later. You do not need to create an account for the examination.
2. Unlock the screen, connect the cable and tap **Trust** when **Trust This Computer?** appears. The iPhone asks for the unlock code.
3. In the tool, on the **iPhone** page, click **Verificar Conexão** (Check connection).

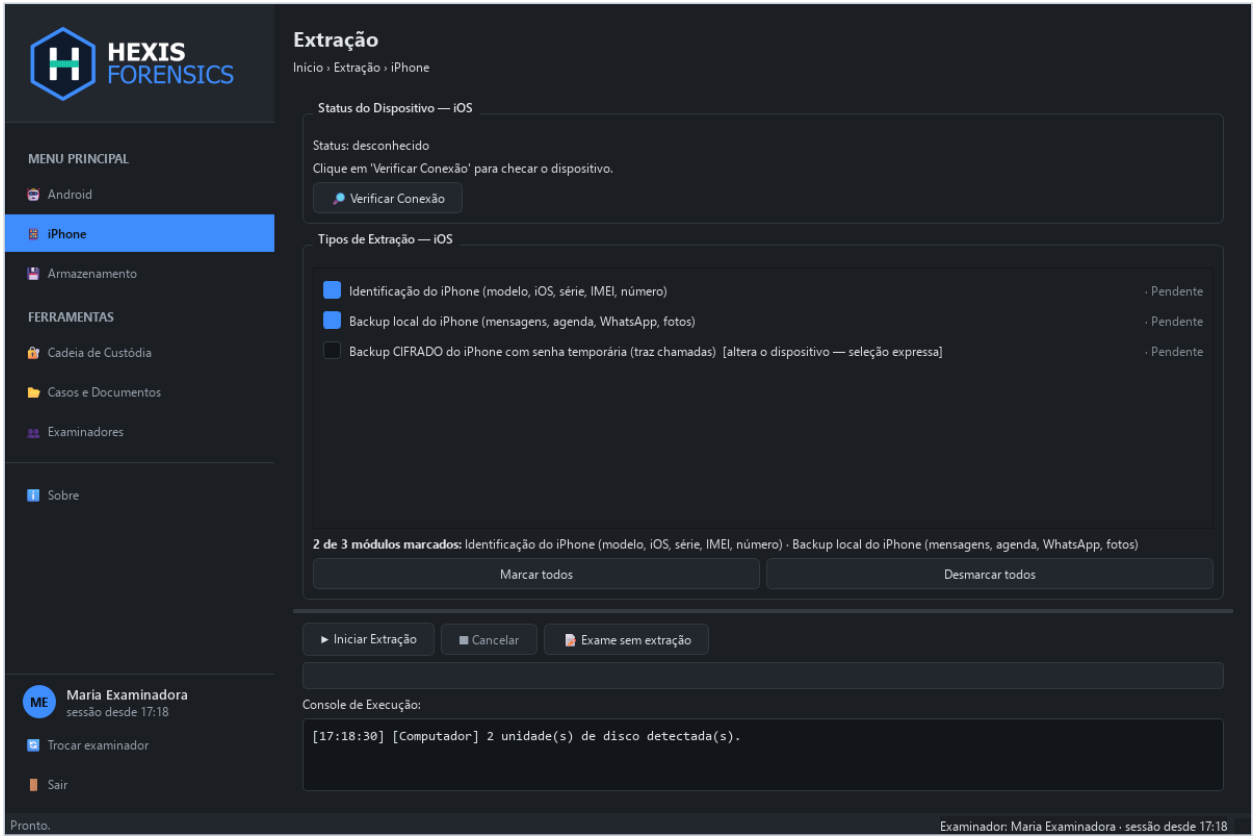


Figure 28 — iPhone page.

> **Warning.** During the backup, the iPhone must stay **unlocked**. With **Low Power Mode** on, auto-lock is stuck at 30 seconds. Prefer to keep the screen active by tapping it. If you need to change **Auto-Lock** or turn off Low Power Mode, that is a change made to the examined device: record it in the examination notes.

The modules

Module	What it does	Alters the device?	Requires
Identificação do iPhone (modelo, iOS, série, IMEI, número)	Lê o que o iOS declara de si mesmo: modelo, versão, número de série, UDID, IMEI, número de telefone e ICCID. Não escreve nada no aparelho.	yes — O iPhone registra a estação como computador confiável (pareamento).	profile DEFR
Backup local do iPhone (mensagens, agenda, WhatsApp, fotos)	Faz o backup local do iPhone pelo mesmo protocolo do iTunes, sem alterar a configuração do aparelho. Traz iMessage/SMS, agenda, fotos e dados de aplicativos (WhatsApp incluso), interpretados depois no Relatório de Conteúdo.	yes — O iPhone registra a estação como computador confiável (pareamento) e anota a data do último backup local.	profile DEFR
Backup CIFRADO do iPhone com senha temporária (traz chamadas)	Define no iPhone uma senha de backup gerada pela suíte, faz o backup cifrado — que inclui o registro de chamadas — e retira a senha ao final. ALTERA A CONFIGURAÇÃO do aparelho durante a coleta; o iPhone pede o código de desbloqueio na tela.	yes — Uma senha de backup, gerada pela suíte, é definida no iPhone durante a coleta e retirada ao final; a retirada é conferida.	express selection (starts unchecked); profile DEFR

Common backup or encrypted backup

The common backup does not touch the device's configuration and brings iMessage and SMS, the address book, the camera roll photos and app data (WhatsApp included). But iOS only includes the **call log** and other sensitive data in the **encrypted** backup. The temporary-password module sets a password generated by the tool, makes the encrypted backup and **removes the password** at the end, verifying the removal.

Comparison	Common backup	Encrypted backup (temporary password)
Alters the device?	No (only the pairing, which every backup requires)	Yes: sets and removes a backup password, with a record
Messages, address book, WhatsApp	Yes, read directly	Yes, after decrypting
Call log	**No** — iOS does not include it	Yes, after decrypting
Photos and their GPS	Yes	Not read in this version
Code on the iPhone screen	Once (trust)	Up to three times

Both can be checked together: each backup goes to its folder (**backup_ios** and **backup_ios_cifrado**), and the content is copied twice. In the encrypted backup, the iPhone asks for the code **up to three times**: to accept the temporary password, sometimes during the backup and to remove it. Keep the device at hand.

> **Warning.** If the owner has already set a backup password, the temporary-password module **does not run** — changing the owner's password would be an irreversible change. Use the common backup: it comes out encrypted with the owner's password, and the content can only be read with it (Decifrar backup button).

Decrypt the backup

1. Open **Casos e Documentos** (Cases and Documents), choose the case and the iPhone's acquisition.
2. Click **Decifrar backup** (Decrypt backup).
3. When the backup was made with the temporary password, the password **comes already filled in**: the tool reads it from the acquisition's **senha_do_backup.txt** file, and only if it matches the seal. Confirm.
4. The decrypted databases go to the case's **decifrados** folder, outside the sealed acquisition, each with its provenance alongside. The password is recorded in the case addendum, so that another examiner can repeat the decryption.

The password also appears in the collection outcome and in the console. If the backup was encrypted with the owner's password, that is the one you type.

What appears in the Content Report

- Each block says which backup it came from: **common iPhone backup** or **encrypted iPhone backup, decrypted**.
- An encrypted backup not yet decrypted: the report says that **nothing is asserted** about the content — not that it is empty.
- A file iOS did not put in the backup (the calls, in the common backup): the report says the backup does not contain it, and that this does not prove its absence.
- **WhatsApp**: texts, photos, audio, groups and shared locations, with the coordinates and the **open on the map** link. Records that WhatsApp itself generates — the encryption notice, the creation of a group — appear with the **System** sense and the note “not a typed message”. A type of record the tool does not know appears as **not recognized**, with no guess about what it is.
- **Photo locations**: the camera-roll photos with GPS (see chapter 9).

07 Examination without extraction

A device locked by a credential, that does not turn on or that is damaged: the examination exists even without extraction, and the negative expert report needs the same chain of custody.

1. On the Android or iPhone page, click **Exame sem extração** (Examination without extraction).
2. Follow the same case, authorization, item, seal and photograph screens.
3. On the form, state whether the device turns on, the condition in which it was received, the battery, whether it was isolated from the network, the **type of lock** and what was attempted. The recommendation comes already filled in.
4. If the device turns on, the tool **attempts the connection without altering it** and keeps the response as evidence: on Android, the ADB listing (for example, “NOT AUTHORIZED”); on iPhone, a session with no pairing request.
5. Generate the **Expert Report** in Cases and Documents: the “Examination without Extraction” version comes out.

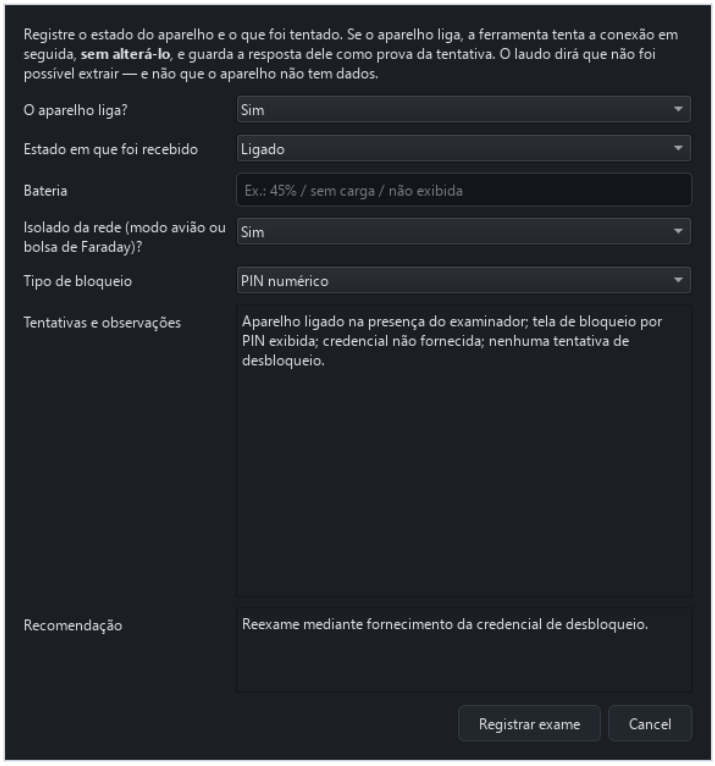


Figure 29 — Device condition and attempts.

The conclusion states that extraction was not possible, and why — and that this **does not** allow any conclusion about the content of the device.

08 Disks and USB drives

The Storage platform examines drives connected to the computer: USB drives, external disks and the disk removed from a computer under examination.

Warning. The disk of a computer under examination is **removed** and connected to the station through a dock or adapter, with a hardware **write blocker** between the two. The tool does not replace the blocker. The examiner's own station appears in the list, identified, so it is not chosen by mistake.

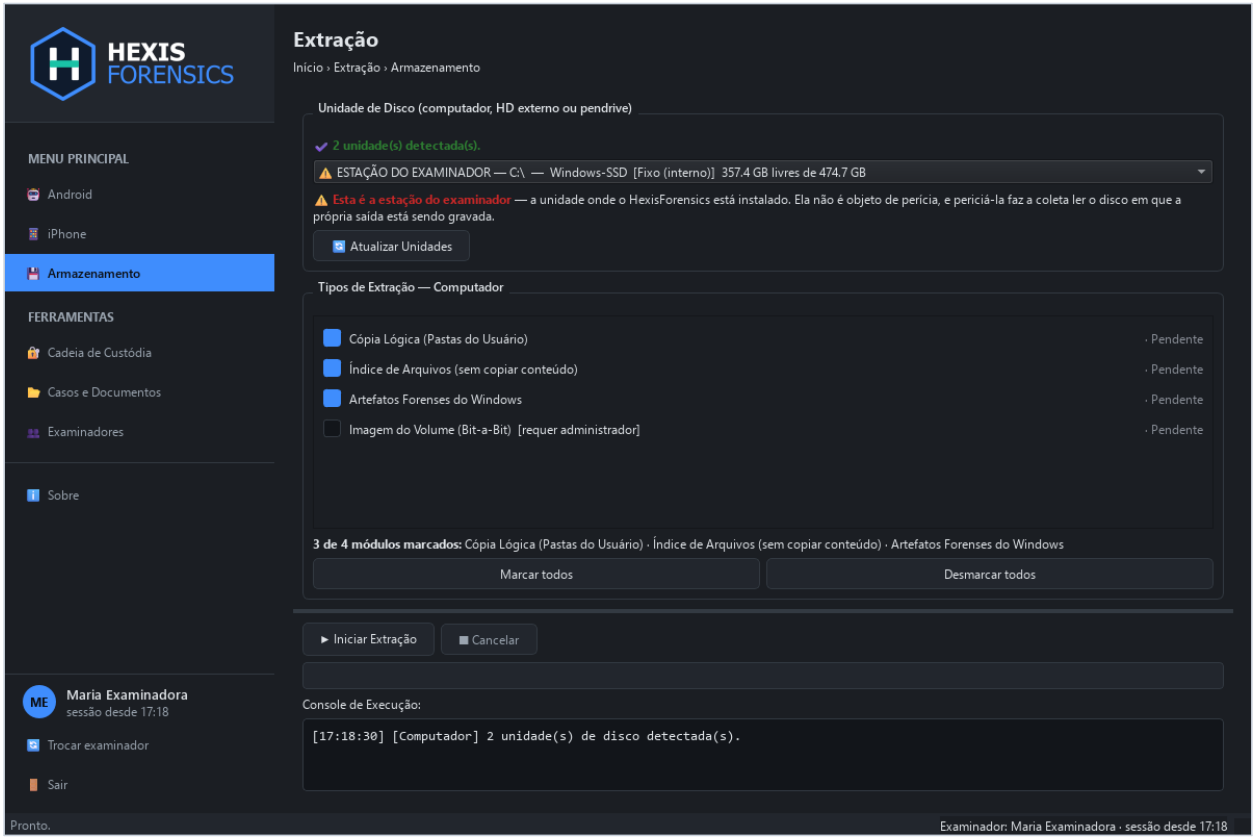


Figure 30 — Storage page: choosing the drive.

Module	What it does	Alters the device?	Requires
Cópia Lógica (Pastas do Usuário)	Copia as pastas-chave do usuário (Documentos, Área de Trabalho, Downloads, Imagens) encontradas na unidade selecionada, com hash SHA-256 por arquivo para a cadeia de custódia. Não requer privilégios especiais.	no	profile DEFR
Índice de Arquivos (sem copiar conteúdo)	Varre toda a unidade e gera um índice (CSV) com caminho, tamanho, datas e hash SHA-256 de cada arquivo, sem copiar nenhum conteúdo. Pode demorar bastante em unidades grandes.	no	profile DEFR

Module	What it does	Alters the device?	Requires
Artefatos Forenses do Windows	Coleta hives de registro, Logs de Eventos, Lixeira, histórico de navegadores (Chrome/Edge/Brave/Firefox), acervos de correio eletrônico (Outlook, Thunderbird, Correio do Windows) e Prefetch da unidade selecionada. Pressupõe que a unidade contém uma instalação do Windows.	no	profile DEFR
Imagem do Volume (Bit-a-Bit)	Gera uma imagem bit-a-bit (.img) do volume selecionado, lendo o dispositivo bruto pela API do Windows, com SHA-256 calculado durante a aquisição e conferência do tamanho ao final. Requer privilégios de Administrador.	no	requer administrador; profile DES

The **Volume Image** requires opening the tool as Administrator. A volume encrypted with BitLocker is detected and reported.

09 Cases and Documents

Everything you do after the extraction: documents, examination of the material, delivery and organization of the cases — without having to open the folders in Windows.

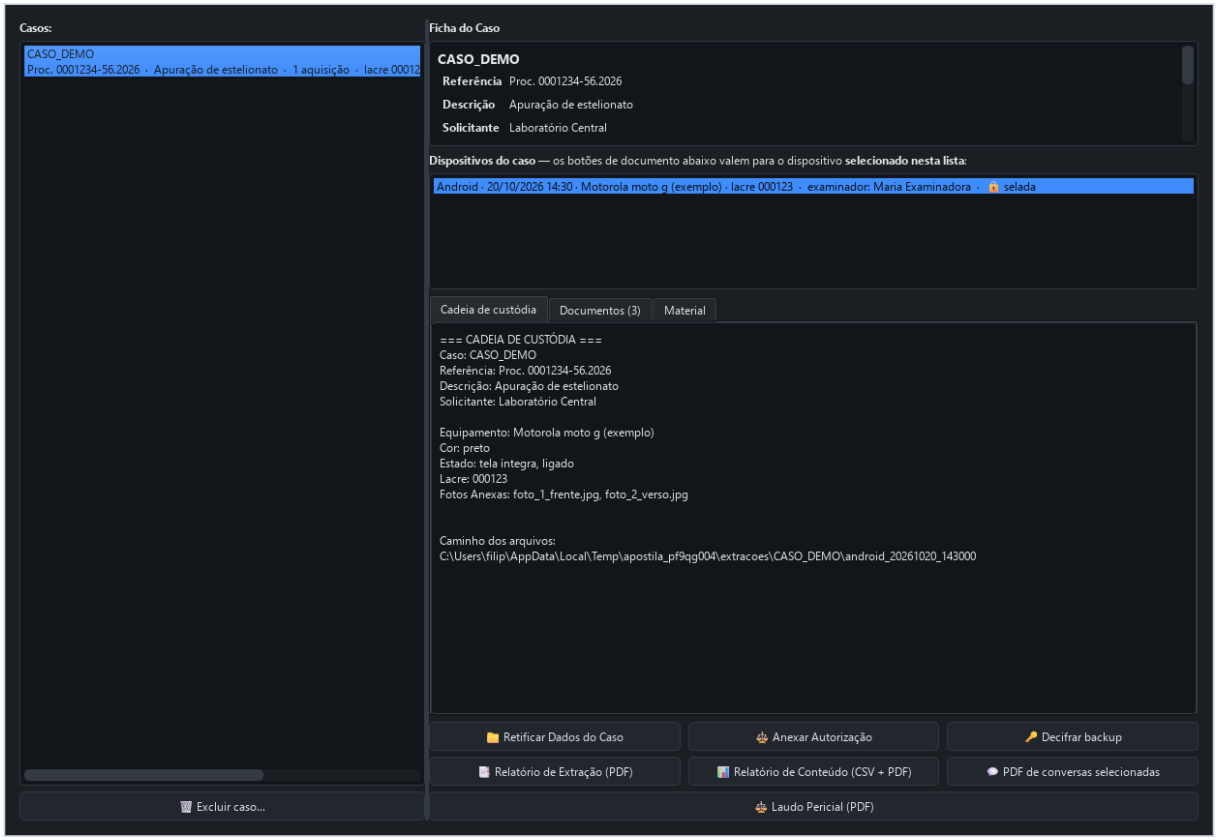


Figure 31 — Cases and Documents: cases, record, devices and actions.

The documents

Document	What it is for
Extraction Report	Timeline of the procedure, environment, modules, outcome and the hash manifest.
Content Report	The messages, calls and contacts interpreted: CSV spreadsheets, HTML notebook and PDF, with the origin of each line.
PDF of selected conversations	Only the relevant conversations, complete.
Expert Report	The final document, signed by the session's examiner.

The documents open in the **tool's viewer** (pages, zoom, print, save a copy) and are listed on the **Documentos** (Documents) tab. They are written **outside** the sealed acquisition, which is never altered.



Figure 32 — PDF viewer.



Figure 33 — The first page of an expert report.

Photo locations

The Content Report brings the section **Photo locations (GPS recorded in the images)**: one line per photo that has coordinates in the EXIF — the data the camera records inside the file. It applies to Android JPEG and iPhone HEIC.

Column	What it is
Date and time	That of the **clock of the device** that took the photo.
GPS time (UTC)	That of the **satellites**, recorded with the coordinate. It does not depend on the device's clock: if it was wrong, this is the reference.
Latitude, longitude, altitude	Where the photo was taken, according to the EXIF.
Camera	Maker and model recorded in the file.
Map	**Open on the map** link, clickable in the PDF.

- Only photos that **match the seal** are included. A photo altered after the acquisition is rejected, with the reason.
- The photographs the examiner takes of the item (front and back) are **not** included: the examination table is not a place where the device has been.
- The same photo copied by two modules counts only once.
- A **0,0** coordinate — typical of a camera with no GPS signal — does not become a location: it goes to the discards, with the reason.

Warning. The coordinate says where the **photo** was taken, not where the device was. The EXIF is editable, and a photo received from someone else carries the location of their device. Messaging apps usually strip the EXIF: a photo with no GPS does not prove the device does not record location.

PDF of conversations

The **PDF de conversas selecionadas** (PDF of selected conversations) button uses the acquisition marked in the list. The window title shows the case and the acquisition: if the list comes up empty, first check that it is the right acquisition.

Examine the material: the Material tab

The **Material** tab shows the acquisition's folders and files, with the preview (photo, text, spreadsheet, PDF, database tables) and the **check of the SHA-256 against the seal**: matches, DIVERGES or not listed. Everything is read-only.

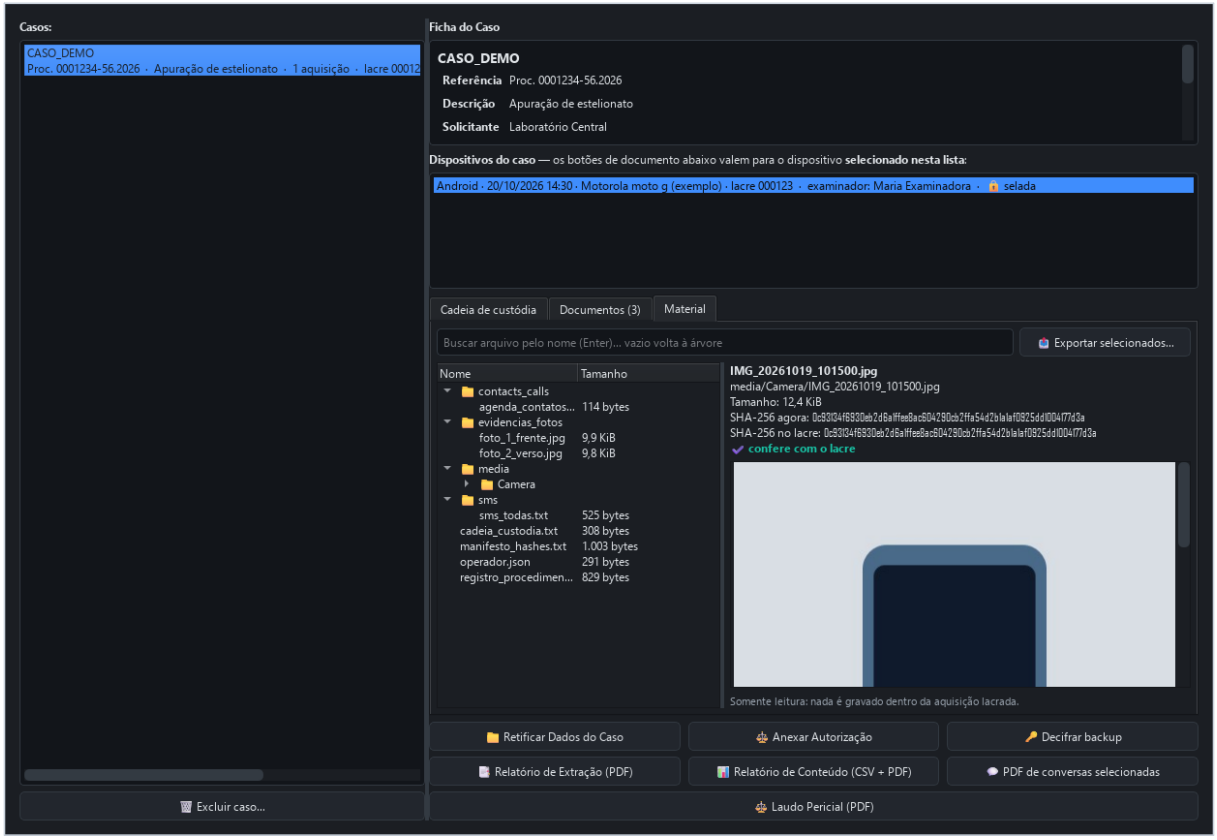


Figure 34 — Material tab.

> **Tip.** Prefer the Material tab to Windows Explorer. While browsing, Explorer can create hidden files (Thumbs.db, desktop.ini) inside the sealed folder, and the check then flags an added file.

Deliver: Export for delivery

1. On the Documents tab, select documents — or, on the Material tab, files and folders (Ctrl and Shift for several).
2. Click **Exportar para entrega** (Export for delivery) or **Exportar selecionados** (Export selected).
3. Choose the destination (USB drive, folder) and write the purpose or the recipient.
4. The tool copies, checks each copy by its SHA-256 and generates the **delivery receipt** (PDF and text), with signature fields. The delivery is recorded in the case.

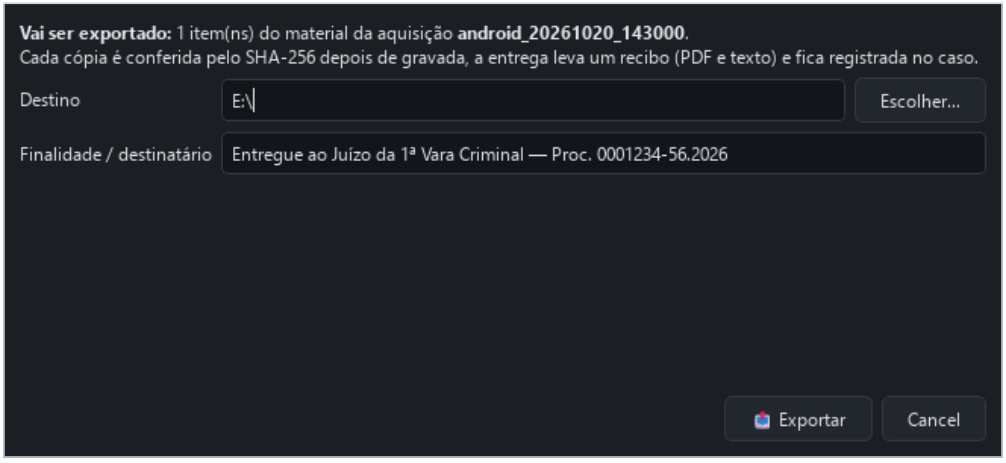


Figure 35 — Export for delivery.

Amend, attach, delete

- **Retificar Dados do Caso** (Amend Case Data) and **Anexar Autorização** (Attach Authorization) enter as a case **addendum**: the original is preserved and what changed is recorded.
- **Excluir caso** (Delete case) requires the case name to be typed, refuses a case with an acquisition still unsealed, and sends everything to the **Windows Recycle Bin** (restorable). The deletion is kept in the access log.

10 Chain of custody and integrity

How the tool proves the material has not changed since collection.

The seal of each acquisition

At the end of the extraction, the **manifesto_hashes.txt** file receives the SHA-256 of each acquired file. The hash of the manifest itself appears in the footer of every page of the documents. Altering any file, or the manifest, changes that number.

Check the integrity

In the side bar, **Cadeia de Custódia** (Chain of Custody) checks an acquisition against its manifest and records the result of the check (no check erases another).

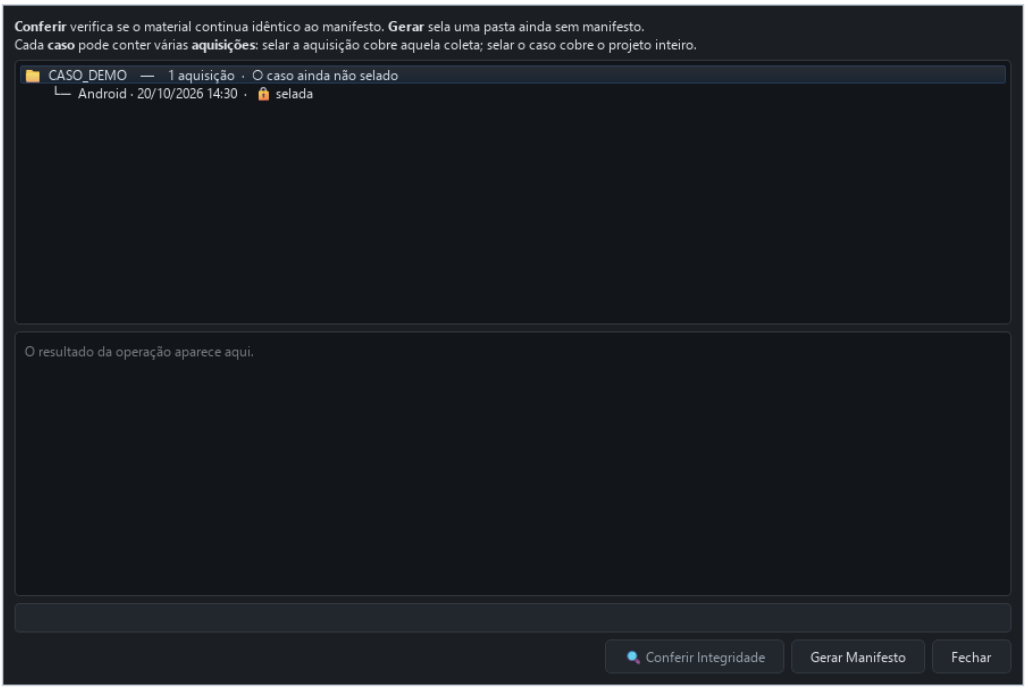


Figure 36 — Integrity check.

Where each thing lives

Case folder	Content
<platform>_<date>_<time>	The sealed acquisition. Never rewritten.
documentos	Extraction Report, Expert Report, Conversations.
relatorios	Content Reports (CSV, HTML, PDF).
decifrados	Decrypted backups, with their provenance.
complementos	Amendments, authorizations, decryptions, deliveries.

11 Tested devices and limitations

The tool states what it verified and what it did not verify — including about itself. This is the list of devices on which version 1.0 was proven, and of the known limitations.

Where version 1.0 was proven

Device	System	What was proven
Motorola moto x4	Android 9	Full extraction, WhatsApp via the 64-digit key, photos with GPS
Samsung Galaxy A22	Android 13 (One UI 5.1)	Full extraction, agent app, shifted-clock alert
Android emulator	Android 13, with elevated privilege	Modules that require elevated privilege, WhatsApp
iPhone 11	iOS 18.6.2	Common and encrypted backup, decryption, WhatsApp, photos with GPS
USB drive (7.28 GiB)	Windows 11	All four modules, including the volume image

Other models and versions tend to work by the same route, but were not verified. When a module fails on a device, the tool records the refusal and concludes nothing about the content.

Known limitations

- **Xiaomi, Redmi and POCO (MIUI / HyperOS)**: the agent app was not verified. These devices usually require “Install via USB” and a Xiaomi account; the install may be refused. The limitation is stated in the module itself.
- **Android 8 or earlier**: out of scope for this version. On a Galaxy J5 Prime with Android 7, the File System Dump stopped early in the copy of the internal storage and still showed “Completed”; WhatsApp Business in the old location is not searched. On a device like that, check each module's reconciliation or use another tool.
- **Android 12 onward** no longer has the system backup: SMS and calls are read directly; when the system refuses that reading, they come in through the agent app, which is express selection.
- **IMEI**: from Android 10 onward, the system does not deliver it without elevated privilege. The tool records each route attempted and declares that it did not obtain it.
- **WhatsApp on Android**: only the backup protected by the 64-digit key (crypt15) is decrypted. The WhatsApp downgrade module has not yet been tested on a real device.
- **iPhone**: the photos from the encrypted backup are not read in this version (those from the common backup, yes). With a backup password set by the owner, only the common backup, decrypted with the owner's password.
- **Out of reach**: device unlocking, passcode bypass, physical extraction of a phone without elevated privilege and recovery of deleted data.

12 Common problems

What to do when something does not go as expected.

The device shows as “not authorized”

Unlock the screen and accept **Allow USB debugging**. If the question does not appear, turn USB debugging off and on, or revoke the authorizations in the Developer options and connect again.

No device responds

Change the cable (it may be charge-only), set the USB mode to “File transfer” and check that USB debugging is on.

The device refused the agent app

See whether there is a confirmation on the device's screen (Play Protect). On Xiaomi, turn on “Install via USB”.

The console warns that the device's clock is shifted

The device was showing a different date and time (common on a device with no SIM or no internet). Do not correct the clock: the measured offset is recorded, and the Extraction Report and the expert report warn that the dates recorded by the device should be read with that correction.

Windows showed “Windows protected your PC”

It is the warning for a program with no paid digital signature. Check the installer's hash, click **More info** and **Run anyway** (chapter 2). If there is no button, Smart App Control is on and requires a digital signature: install on another computer, ask your IT department to allow the program, or run from source (chapter 2).

The tool does not find the iPhone

Check that the Apple Devices app (or iTunes) is installed, **disconnect and connect the cable again**, unlock the screen and tap **Trust**.

The iPhone refused the backup (code 212)

The device's initial setup was not completed. Finish it — the Apple account can be skipped — and try again (chapter 6).

The iPhone backup stopped halfway

The screen locked or the cable moved. Keep the iPhone unlocked (watch out for Low Power Mode) and type the code whenever it asks.

The conversation list came up empty

Check, in the window title, that the acquisition is the right one. If it is, the Content Report says why there is no interpreted conversation.

The report says “the provider responded and returned no record”

Android itself answered the query that the source is empty — for example, SMS on a device with no SIM. It is not a read failure. Deleted records do not appear by this route.

A file appears with “(nome alterado 1)” (name altered 1)

Windows does not accept the name the file has on the device (two names that differ only in case, or a trailing dot). The content is the same, checked by the hash, and the correspondence is in the reconciliation report.

WhatsApp shows “On · Password”

The backup is protected by a password, and the tool decrypts only with the 64-digit key. Turn off the protection and turn it on again with the key (chapter 5, steps 1 and 2).

The recent WhatsApp messages do not appear

The backup predates them. See the “Source note” in the report and make a new backup before extracting.

I forgot the password and the code

Create a new profile. The old acquisitions stay under your name.

13 Glossary

Terms used in the tool and in this manual.

Term	Meaning
ADB	Android Debug Bridge, Android's debugging tool.
Acquisition	An extraction from a device, written to a sealed folder inside the case.
Case	The set of acquisitions of a single examination.
Addendum	Information added to the case later, without rewriting the acquisition.
Encrypted backup (iPhone)	A local backup protected by a password; it is the only one in which iOS includes the call log.
crypt15	The format of WhatsApp's encrypted backup.
EXIF	Data the camera records inside the photo: date, camera and, sometimes, GPS.
GPS (UTC)	The time of the satellites recorded with the coordinate; independent of the device's clock.
DEFR / DES	First responder and specialist, in ISO/IEC 27037.
Hash (SHA-256)	A file's fingerprint: any change alters the value.
Seal (manifest)	The list of hashes of all the acquisition's files.
Express selection	A module that alters the device and only runs if chosen one by one.
Caveat	What the module did not reach, declared alongside what it obtained.